Monitra: AI Based Study Habits and Posture Monitoring

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ABSTRACT

In the present, both parents in a family are often needed to work for a better financial lifestyle, and their children are often either sent to after-class care or stay at home. Often times, children are expected to self-study if they were to be left alone at home, by completing their homework or going extra mile on additional study. However, children may not follow through with their self-study sessions and instead of having a balanced schedule or a discipline time management, children may neglect their studies. This can be due to several factors such as lack of motivation, lack of encouragement or environmental factors and traditional supervision such as monitoring by parents has been made impossible in today's financial economic state. Harsher methods such as reducing their distraction like confiscating devices are not practical due to the importance of communication in case of emergencies or related actions. This results in parents being unable to supervise or provide the encouragement or motivation needed by children due to them being unknown to their children's study habits. Additionally, another issue that comes along with studying is their posture health. It has been proven by many researchers that prolonged sitting will causes posture health issues regardless of their age. Hence, Monitra aims to address these 2 issues at once, by assisting parents in monitoring and informing them about their children's study well-being while maintaining privacy and preventing posture health issues at an early age. In this project, Monitra's primary technology is to use Visual-Language Model, VLM, to monitor children's studying activities due to the ability of VLM to understand the contextual meaning of video frames for reporting purposes compared to traditional method such as Computer Vision. Posture detection will also be processed by using pre-made models that are lightweight for a better real-time reminder.

Area of Study (Minimum 1 and Maximum 2): Artificial Intelligence, Mobile App development.

Keywords (Minimum 5 and Maximum 10): AI, VLM, LLM, Posture Detection, Mobile Application, Activity Monitoring

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AI	Artificial Intelligence
COSM	Children On-Screen Monitoring
VLM	Visual-Language Model
LLM	Large-Language Model
CV	Computer Vision

CHAPTER 1

Project Background

In this chapter, the project background will be presented, along with motivation which drives the need to create Monitra, and contributions, what this project has to offer to its targeted users.

1.1 Introduction

As technology has made its advances, things people do in daily lives are made easier, and it ranges from automating tasks to assisting tasks. In this project, the main focus is assisting user's tasks, more precisely, a small academic and posture health assistant for children while informing their guardians of related information. In Malaysia, children are referred to individuals under the age of 18 [1], and in the present, parents are often busy with their work schedules and are unable to monitor and encourage their children at home, especially academically, leading to parents being uninformed about their child's education issues[2]. This is indeed very true in today's economic state where both parents often need to work to support a decent and comfortable life, unlike traditional times where only one parent, usually the father, to be the family's sole breadwinner. Hence, the absence of attention and encouragement causes children to have low motivation and perform poorly academically[2]. Therefore, technology can play a supportive role in this matter, whereby a technology assistant can monitor children's study activity at home. This, in turn, can help parents to stay informed about their children's studying activities, enabling parents to have easier time supporting their children during their available time. Moreover, in the domain of having study sessions, posture health is an important issue to be addressed. This is because individuals' spinal health can be affected due to prolonged sitting. Studies have shown that sitting for long hours to study or usage of computers for long hours can affect one's posture health. In a study, their findings show that student's posture health and comfortability are affected by long hours of sitting when attending classes in university[3]. It is also mentioned by another author in the study that prolonged sitting causes individuals to have bad posture and discomfort among themselves[3]. Hence it is important for children to be aware of posture health issues, since early awareness can benefit children where they can have better practice of sitting posture during study sessions. This is where technology such as this project comes into play, where it can not only monitor children's studying activities and inform their guardians, but it also aims to assist children's posture in real-time, reminding them if their posture is improper.

1.2 Problem Statement

In the present with advance technologies, children are often distracted during self-study, reducing their study efficiency. This may be caused by poor supervision, poor environment, low motivation or external distractions. Traditional supervision, such as confiscating phone, are not practical anymore, as it is essential for safety and communication. Although parents prefer to monitor their children's study habits, work schedules or privacy concerns often prevent them from doing so. Hence, without monitoring, children may not follow through with their study session. Additionally, whether it's long or short study sessions, children may have poor posture which would negatively impact their spinal health, especially since they may lack ergonomic support.

1.3 Motivation

To tackle the issues such as distraction during self-study which may be due to poor supervision or poor environment, this project aims to let their guardians stay informed generally of their activity. Moreover, to address the posture health issue faced by children during study sessions, this project also aims to help provide reminder to the children to encourage better posture health. Monitra will be designed to act as an assistant as well as a guardian when it comes to monitoring children's self-study sessions and helping with their posture health at an early age. It goes without saying that Monitra is a product mainly for monitoring children, which is the self-studying activity and informing guardians of the related information without compromising privacy. This helps guardians that are not able to be around children to at least accurately know what their children are up to generally while giving them privacy. By allowing guardians to stay informed of their children's study activities and promoting good posture health, Monitra hopes to provide children with more focus and discipline study environment while maintaining a good body posture.

1.4 Contributions

By the end of this project, there should be a minimum functional product, Monitra, to be use as a monitoring device that detects if children are studying or not along with the aim to improve and help children with their posture health while studying. This provides a monitoring function for children to keep parents informed generally of their study activity and keep track of their posture for better spinal health by providing real time response. To inform parents, a mobile application will be provided which will be used to show a user-friendly report and dashboard on how much time their children spend on studying or activities that they are not studying. This allows an easy way for parents/guardians to stay informed of their children's activities in general without compromising their privacy. Parents that are busy and unable to watch over their children will be able to at least know what they are up to at home regarding their study behaviour. Moreover, Monitra will be able to help remind children of their posture when studying in real-time, allowing children to be aware of their posture health and actively correct their sitting posture such as head, neck or their spine.

CHAPTER 2

Literature Review

Currently, there is none existing technology or product that specifically monitor both children studying activities and posture at the same time. However, plenty of methods have been explored and developed in their own domain, such as monitoring children's activities in different ways, and posture monitoring that ranges from camera sensors to wearable devices.

2.1 Existing work on children monitoring

In the context of existing method of monitoring children, it includes tracking children behaviour digitally, parental control, different software that uses AI and tracks children studies behaviour, to manually monitoring devices such as using Wi-Fi camera.

2.1.1 Children On-screen Monitoring Application

Children On-screen monitoring application, COSM, was created by a graduate student in UTAR where it was able to track children activity only during online class sessions[3]. To do so, the system will check user's devices for E-learning applications such as Microsoft Teams, Zoom, Google Meet, and other related applications, and if any of the applications exist, the system will then be put to work[3]. This system does not use any AI to achieve their objectives and instead, it has functionalities such as on-screen monitoring, browser activity tracking, entertainment activity tracking and even methods such as keylogger and mouse tracker. The system will then send alerts to via Telegram app to inform parents[3]. Hence, this software created were able to alert parents of children's activity digitally and only during online class sessions.

The strength of this software is that, firstly is has feature that tracks user's mouse movement. As mentioned by the author, the mouse tracking feature is a module that will have a quick report to the parent when the mouse is idle for a set amount of time, indicating that the child may not be active during online class session. Furthermore, the module will also produce few beeping sounds to act as a reminder for children to be active in class should they loses focus or distracted. This feature works well to assist children maintaining a more consistent focus during online class, since children may be present in online classes, but are not concentrating behind the screen. The next strength is the software's keylogger module, where it records whatever keys are pressed on the keyboard in a text file which can be review by parents[3]. This further monitor their children during online class session, where it gives parents more clarity of what their children were doing even if they are active in online classes.

However, limitations exist in this project, since the software are only able to be utilized during online classes, monitoring of children's studying activities are not possible if children were to conduct self-study sessions without being in an online classroom. Online classes are conducted frequently during quarantine time to tackle Covid-19 pandemic, and with the pandemic tackled, this software of monitoring children is not as effective anymore. Furthermore, as mentioned, it does not effectively monitor children study activity, for children may still be studying even without joining an online class. Additionally, the project method of informing parents is via Telegram that sends alerts, reports, and even text files but this isn't user friendly for it lacks a good user interface for accessibility.

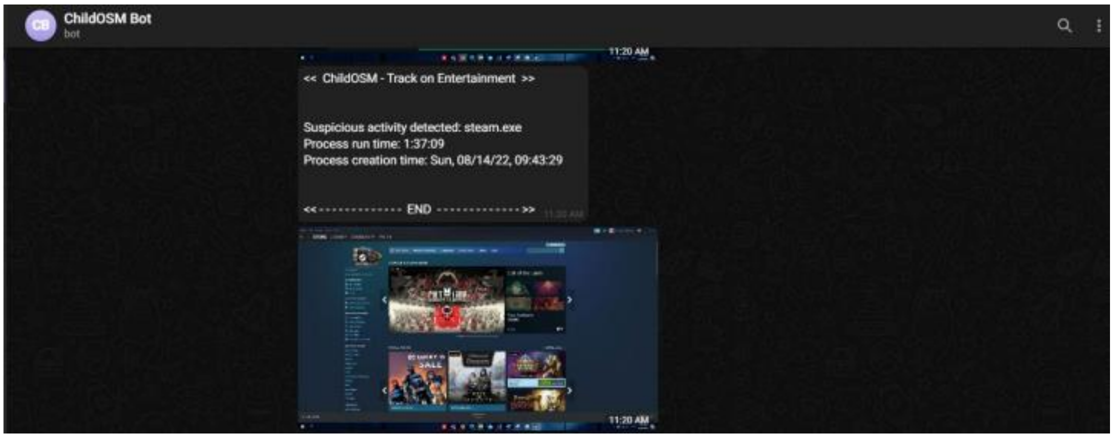


Figure 2.1, Example of report via Telegram by COSM

2.1.2 Qustodio

Qustodio is a parental control app that monitors children activity digitally instead of just focusing on studying activity. The app works across different devices such as Android, Mac, Windows, and even in iPhone and iPad. It has 2 apps where one is for parents and one for targeted device, whereby in the parent account, users can add multiple child account and monitor their digital activity. Qustodio has features that effectively monitor and manage children's account activity [4], and that includes **(i)** filtering content & apps, **(ii)** monitoring activity such as browser history and activity timeline, **(iii)** setting screen time limits or pausing Internet connection, **(iv)** a GPS tracking that allows parents to know their whereabouts, and **(v)**, a feature that tracks calls and SMS of the targeted device. Additionally, AI-powered alerts were implemented to send reports and alerts to parent's app where the AI automatically notify the parents of any suspicious activity or harmful content.

With all these features, Qustodio works very well in informing parents of their children's behaviour digitally and controlling children's activity digitally. This is especially important in this technology era where children spend majority of their time using their devices. However, even with its extraordinary features when it comes to monitoring children activity, it has the same limitation as CSOM, where it is not able to monitor children offline study activity. This is because Qustodio only aim to monitor children's activity digitally and not the physical activity of children. It lacks the ability to monitor children's activity physically aside from tracking their location. Furthermore, Qustodio raises privacy concerns due to its features of monitoring calls and messages of children, as well as accessing browser history. Although it is a strict way to monitor their activity digitally, this would make children to feel insecure as if their actions are constantly monitored which is not recommended for children's wellbeing.

2.1.3 CuboAI

CuboAI, a monitoring system that comes with their own camera hardware, with a purpose to monitor babies' activity and provide alerts when triggered by crying or entering restricted zone specified. Although the work done here is not monitoring children studying activities, it is also worth taking note of their ability to monitor babies, by using their own camera, video frames were captured and sent to their own cloud storage and perform their own AI processes features. The features include detecting baby's covered face, cry detection, danger zone coverage, breathing analysis and temperature detection[4]. This undoubtedly shows that it performs very well in their own niche of monitoring baby using AI to assist parents. Moreover, this product comes with their own camera mobile stand and floor stand to place the camera, as well as CuboAI Smart Baby Monitor, Smart Temp for temperature detection, and Sleep Sensor Pad.

In order for parents to view playbacks of longer time in the past, a payment is required meaning the review time of recorded video are limited and needs a paid cloud subscription. Additionally, despite having great features, privacy issue arises on how the product handle the recorded video and information. Recorded video are needed to be sent to their cloud storage for processing but the issue of privacy remains here whereby their data are being sent over to their cloud storage, and the vulnerability of data breaches or related security risks exists. Furthermore, even with having a great security to protect their data digitally[5], the issue of how parents are unable to control how long their data is stored points out that privacy issue is still a concern here, especially for families that are more sensitive about their privacy.

2.1.4 Existing work on posture monitoring

Posture monitoring has been explored in various methods that ranges from hardware combination such as wearable devices that provides reminders or ergonomic chair with software, to purely software monitoring using AI.

2.1.5 UprightGo

UprightGo is a wearable device that monitors user's posture by using biofeedback science. It comes with a small product that can be wear like a necklace behind their back or using a provided adhesive to stick on user's back[6]. The main part of the product is that it contains 2 enhanced gyroscope sensors to detect bad posture and provide a real-time reminder by vibrating the device[7]. Hence, if it were to sense bad posture of user's back such as slouching or other related back bad posture, it will produce vibration that helps remind users of their posture. Additionally, the product also has the capability to store user's posture information in their application, showing the information in a dashboard[7]. This enables the user to lookback on their posture health for up to 30 days and to be aware of their posture health whether it is improving or do they need more efforts and awareness. Auto calibration is also done by the product where it will first recognize user's body posture and perform optimal training on itself about good and bad posture[7] before wearing it throughout the day.

However, despite having a real-time vibration for posture detection, few limitations exist where firstly, the auto calibration and setup of the product can be complex. This is because if users are unable to have a good starting position of their posture to let the product know what is good and what is bad[8], a poor setup will cause the product to not function optimally. Hence, the feedback of using vibration will not be triggered if the setup was wrong ultimately affecting the product's functionality. Not to mention, many users who bought the product stated that it is not useful due to this reason. Secondly, by wearing a device behind user's back all day, this raises concerns about their comfortability. Using adhesive or a necklace like product to place the product behind their back, it will still feel uncomfortable to a few degrees. This is due to having an electronic device that produce vibration, sticking on user's back the whole period of using it. Thirdly, the product only measures bad posture of upper back of the user, by placing the product in between shoulder blades to monitor slouching and so on. This results in correcting only the user's upper back posture, neglecting other posture issues such as leg-crossing or neck tilt.

2.1.6 PostureMinder

PostureMinder, unlike UprightGo, is a simple software based to address posture issue by sending reminder. It is an extension that can be downloaded from Google and provides reminders on interval that would remind users to be aware of their posture. While it has no features that monitor user's posture, it is still worth including as one of the works done for existing solutions to tackle posture health issues. This is due to its super easy to use and implementation, where users just need to download the extension, configure reminder intervals and users are even allowed to toggle the sound of reminder. Not to mention, it also has a walk reminder, and its interval can be adjusted as well. Although it does not have monitoring features directly, a reminder of posture correction still achieves the objective of posture monitoring, which is to let users have a better spinal health and increase their awareness.

The limitations of this extensions definitely refer to the non-existent monitoring posture, although it works well in what they tend to offer, which is just to provide reminders, it is also way beneficial to directly monitor user's posture instead of providing reminders only especially with existing technology out there. Besides that, the extension of course only works if Google Chrome is running in the background, so it will not be able to provide a 24/7 reminder if Google application is closed, limiting it usage.

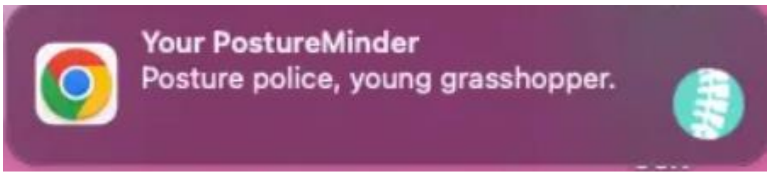


Figure 2.2, Reminder of PostureMinder extension

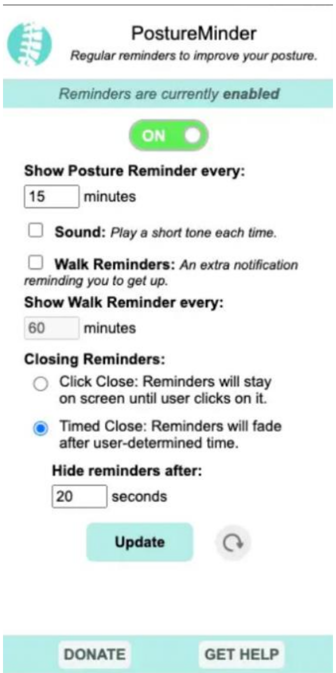


Figure 2.3, Configuration of PostureMinder extension

2.1.7 ZenPosture

ZenPosture is a software-based posture monitoring, that accesses user's webcam and perform posture analysis using AI. This application access user's webcam to capture user's upper body, mainly to tackle posture issue when working[11]. Analyzation of upper body posture using AI is done locally on user's devices and not being sent over cloud to be processed. Like any other posture monitoring software, ZenPosture's way of informing users on their bad posture is similar to PostureMinder, where it uses a lightweight alerting system to provide real time reminders to user[12]. It has been stated that their AI uses TensorFlow to train data and plot points on user's shoulder and nose, to determine bad posture[12]. Moreover, the app also provides personalized insights of user's progress along with statistics and improvement recommendations.

Despite having a unique idea, limitations still exist in their product, whereby the accuracy or effectiveness of the AI during monitoring will be affected by webcam quality. This is due to external environments that may affect the webcam such as potential dirt on the webcam, or lightings of environment, all affecting the accuracy of their AI model. The application currently, does not have any ways such as reminder or informing ways to inform users of potential causes that will affect the quality of monitoring, hence it may result in inaccurate response. Additionally, even though privacy issue may not be of concern here due to the frames of video captured are processed locally, this introduce a new problem, where user's webcam and AI processing are running as long as the application is toggled on, forcing a heavy load of work on user's devices. This in return, may affect user's work productivity that ZenPosture aim to achieve.

2.1.8 Work done on using VLM for monitoring purposes

Visual-Language Model, VLM are popular during 2020 – 2021 due to its ability to understand context of videos or pictures. It has been said that current limitation of present technology, which is Computer Vision, CV, leads to the creation of VLM[16]. It is shown that by implementing VLM, it was able to perform certain task such as understanding context, Question Answering according to the input and datasets that were trained on[16]. This bridge the gap of CV, where it was unable to understand context and only works in object detection. Hence, VLM is something to be researched on to further understand and utilize its potential in various area of today's life and monitoring is also part of this area to be worked on. Although they are no big products of software that uses VLM to monitor types of things yet, various of research has been done and some notable studies will be examined below.

2.1.9 Idefics Model for Driver Monitoring System

This study shows the implementation of VLM to monitor drivers in vehicles. The specific model chosen was Idefics, a model that is suitable for in-vehicle applications[13] and the purpose of this study is to use VLM to understand driver's state in driving scene compared to traditional methods that uses a combination of sensors and algorithms. It was shown that the study uses various method on Idefics to successfully make an accurate response of driver's state, and the methods include Zero-shot prompting, Few-shot prompting, and Prompt-engineering. Although Zero-shot prompting were not accurate, few-shot prompting shows significant results. Meanwhile, prompt engineering case has mix results, where the accuracy are not consistent due to the prediction of multiple variables in one time and the study made an important point that stated for future work, variables could be separated to multiple agents, meaning each agent handles only one variables and passes all the information to a model that can conclude the result, which may increase the accuracy[13].

2.1.10 VLM on Activity Monitoring

A small project was worked on where it utilized BLIP, (Bootstrap Language Image Pretraining), to generate image captions. The flow of its system was by using a Pi camera, the video stream was forwarded to the processing system using Flask. A model is used to extract feature vectors and compares frames for changes and passes those frames to BLIP for caption generating[17]. The results of the captions along with the time of generation was then recorded in a file to realized its output[17]. It is shown that BLIP was able to generate the captions correctly, although the total accuracy was not mentioned. The study also noted that, further improvements such as integration of real-time notification can be done to fully utilize this functionality and even generating reports with these data.

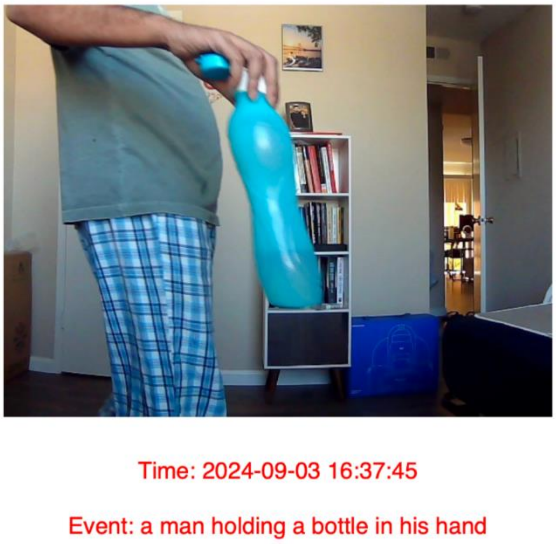


Figure 2.4, BLIP result of image captioning

2.2 Limitation of Previous Studies

Previous studies in the context of monitoring children's study activities have multiple limitations and that includes the inability to monitor children's physical studying activities. This is because both Qustodio and COSM lack the ability to understand what children are doing physically since their features focuses on digital activities. Although, CuboAI does monitor physical activity of babies, privacy issues were the problem due to them storing and processing data on their own cloud. Moreover, in projects like CSOM, it lacks a user-friendly interface that shows reports or dashboards, and instead, reports are being sent via Telegram in text format.

In the context of studies or work done for posture monitoring, products that are wearable such as UprightGo[6], are not comfortable to be wear all day long for every day. Having an electronic device sticking on user's back may not be comfortable despite having a good feature to combat bad posture and contradicts one of the objectives of what we as a product developer aim to achieve, which is to promote a comfortable yet positive environment for posture. Meanwhile, in the software tool such as PostureMinder, does not monitor and get to know user's posture conditions but merely providing reminder, and this leads to not having a report overall for users to know their posture health overall. Application such as ZenPosture has its own limitations where it only manages to correct posture of upper body, limiting the effect of having a good posture. Additionally, ZenPosture uses webcam to monitor user's posture, and this may lead to inaccurate predictions of their AI algorithms due to lighting or related effects. Besides that, the AI processing the image frames were all done locally and this may impact user's device if their device is not able to handle the process, since webcam and processing of video frames happens simultaneously.

In addition to these studies, a VLM research on monitoring also has its own unspoken limitation, which is, in [13], Idefics model were used to monitor driver's state with a correct response, however, this was done on 1 frame each time, and the issue is, one frame of a video is not enough to confirm whether the driver is driving or stopping but the model would still give response that conclude driver was driving along with other related action gotten from the frames.

2.3 Proposed Solutions

This project aims to create Monitra, a **webapp and separate camera** that monitors children's studying activities physically to tackle the limitation of unable to monitor and understand children's studying activities. To achieve this, **Vision-Language Model (VLM)**, will be utilized to capture the context of children's action. In order to address privacy concerns such as CuboAI in [5], video frames captured are to be **processed locally** on user's devices. This would prevent users' data being sent to cloud storage for processing. However, this would introduce new issues such as the impact of performance on user's device due to the local processing of frames captured, similar to the limitations faced by ZenPosture in [12], especially when using heavy models for VLM. Hence the next solution will be implementing **filtering of frames**, whereby a simple and lightweight model will be used to lessen the frames sent to VLM. Depending on the result, the frames will only then be passed onto VLM model for processing the context, preventing unnecessary frames to be processed by VLM every iteration. In addition, this project will introduce a **mobile application** to address the limitation of the lack of user access that is not user friendly. This enables parents to view the activity of children, enabling them to have better user experience by showing dashboards or reports of activity and posture health of their children.

Furthermore, to tackle posture monitoring issues such as the limitation faced by UprightGo which is the comfortability of users wearing a device behind their back, this project solution is the camera that is used for monitoring children activity. By using a **separate camera** to send video frames to an application, this solves the need for wearing devices on the body. Not only that but using a separate camera would also solve the issue of ZenPosture where they use a webcam such as lighting of environment and quality of video frames that could impact the result of VLM. Additionally, frames processed would trigger a **reminder digitally** by using Large Language Model, LLN, according to the results.

Furthermore, when processing frames using VLM, it is crucial to understand the context of what the project aims to achieve in order to not wrongly or inaccurately process frames. As an example, the study of [13] has a limitation of processing one frame and made a conclusion of result and it reflects to this project scenario as well. Posture processing is not to be done by one frame only since users might be adjusting their body or moving around temporarily to take objects or other relevant actions and these actions should not fall into a bad posture categorization. Hence, the solution of Monitra is to **process multiple frames** over a period of time for posture detection to truly confirm the result.

Table 3.1 Limitations and Solutions

Limitations	**Proposed Solution in Monitra**
Inability to monitor children studying physically	Using a separate camera
Privacy issue of user's data	Process video data of users locally on their devices
User's device performance lowered due to heavy resource usage	Implement a lightweight model to only process necessary frames
Lack of user-friendly reporting	Mobile app for parents to view report
Comfortability issue in posture detection	Using a separate camera
Environment affects video quality using webcam	Using a separate camera
Potential of wrong captioning using VLM on just one image	Process several frames over a period of time for concrete results

CHAPTER 3

Project Scope and Objectives

3.1 Project Scope

By the end of the project, the aim is to produce a working software prototype of Monitra. It will contain a web app on targeted computer or laptop that processes video frames sent by a separate camera, along with a mobile application for parents to be informed about their children's activity after being processed from the desktop application. Monitra will have minimum functions as below:

i. Monitor and then classify children's studying activities into True OR False output for reporting.
ii. Implement lightweight models to classify the frames before sending them to VLM.
iii. Frames processed by VLM are to be **stored in a database** for reporting purposes.
iv. A simple yet user-friendly mobile application for parents to have an overview of their children's studying activities.
v. Detecting posture of children when studying and classify output into True OR False for reporting.
vi. LLM to be utilized for text responses as a digital reminder for bad posture.
vii. Dashboard or report to show posture and studying activity information via mobile application.

This project focuses on monitoring activity indoor and only support monitoring one individual for the current stage. Further incremental improvements will be made on Monitra if all minimum functions achieved such as **classifying different types** of studying activities instead of 'Studying (TRUE)' or 'Not Studying (FALSE)', or further improvement for posture health.

3.2 Project Objectives

The objective of creating Monitra is to create a web app on a personal computer that works with a camera to monitor children's studying activities and present its information in a mobile application to inform parents. Posture monitoring is also part of the main objective that works together with the activity monitoring and provides a real time reminder to children. The objectives are as follows:

i) **To tackle privacy issues while monitoring**

As mentioned previously that existing work done on monitoring children activity may have privacy concerns such as processing their info over cloud, Monitra aim to handle this issue while being able to assist parents in monitoring and informing them regarding their children's studying activity.

ii) **To monitor and recognize children's studying activities using VLM**

This project aims to leverage VLM to analyse visual input and response with a better contextual understanding, enabling better results compared to traditional method of monitoring.

iii) **To reduce resources usage during local processing**

Since this project will require user's device to run the frame processing such as the VLM model, lightweight classifier will be implemented to filter unnecessary frame.

iv) **To promote better posture health**

Similar to some of the existing work done on posture monitoring, Monitra current idea of correcting user's posture, is by sending a digital reminder via the application on targeted device.

v) **To inform parents of studying and posture information**

Information gathered by Monitra shall be displayed in a way that is easy to understand for parents in a mobile application.

CHAPTER 4

Methods

The current idea of method to be used in developing this project will be following SCRUM framework from Agile methodology. Although the framework is used usually in a team, implementation of this method as an individual project is not impossible[14] and instead it prevents overwhelming of tasks and managing time. By using this method, the project priorities will be planned out clearly and dividing into manageable chunks of tasks, and by implementing Sprints to complete one task at a time. This will require the individual to manually have a self-review everyday before starting the Sprint and perform retrospective after each Sprint on what could be done better and plan for the next Sprint of the project.

4.1 SCRUM, Agile methodology

4.1.1 Requirements/User stories analysis

The very first phase of this project, which is to clearly think what needs to be completed for Monitra such as reviewing project objectives, breaking it down into clearer task of what to be done.

4.1.2 Project Backlog

Project Backlog will be utilized to list down all specific tasks that are derived from the scope and objectives of the project. Critically analysing what task are to be performed and their respective priorities. Tasks are to be selected one at a time to be worked on during a Sprint and the end result should be recorded. In case of critical issues that causes tasks to unable to be completed by the end of the Sprint, the leftover task should be reviewed and reprioritize in Project Backlog before planning on the next Sprint.

4.1.3 Sprints

Sprints refers to a time duration that are fixed throughout the project development phases, and where one and only one task shall be work on during the Sprint. The duration of Sprint estimated for this project will be **2 weeks** and by the end of every 2 weeks results of task done will be reviewed before moving on to next Sprint.

4.2 System Design

4.2.1 Hardware

Hardware involved in the project includes a laptop to train lightweight AI models, research, and coding of the whole project, as well as a mobile phone that will be used as a webcam by using an app called IP webcam. Reason of using separate camera is to properly monitor children's studying activities and their posture health instead of relying on webcam that only captures a limited amount of body information, which aligns with this project goals that aims to monitor posture health and studying activity.

Table 4.1 Specifications of laptop

Description	Specifications
Model	MSI Laptop
Processor	Intel(R) Core(TM) i5-10500H CPU @ 2.50GHz
Operating System	Windows 11
Graphic	NVIDIA GeForce RTX 3050
Memory	8GB DDR 4 RAM
Storage	477 GB

Table 4.2 Specifications of mobile phone

Description	Specifications
Model	OPPO A9 2020
Processor	Qualcomm Snapdragon 665
Operating System	Android 11
Graphic	Adreno 610
Memory	8GB RAM
Storage	128GB

4.2.2 Software and related models for AI

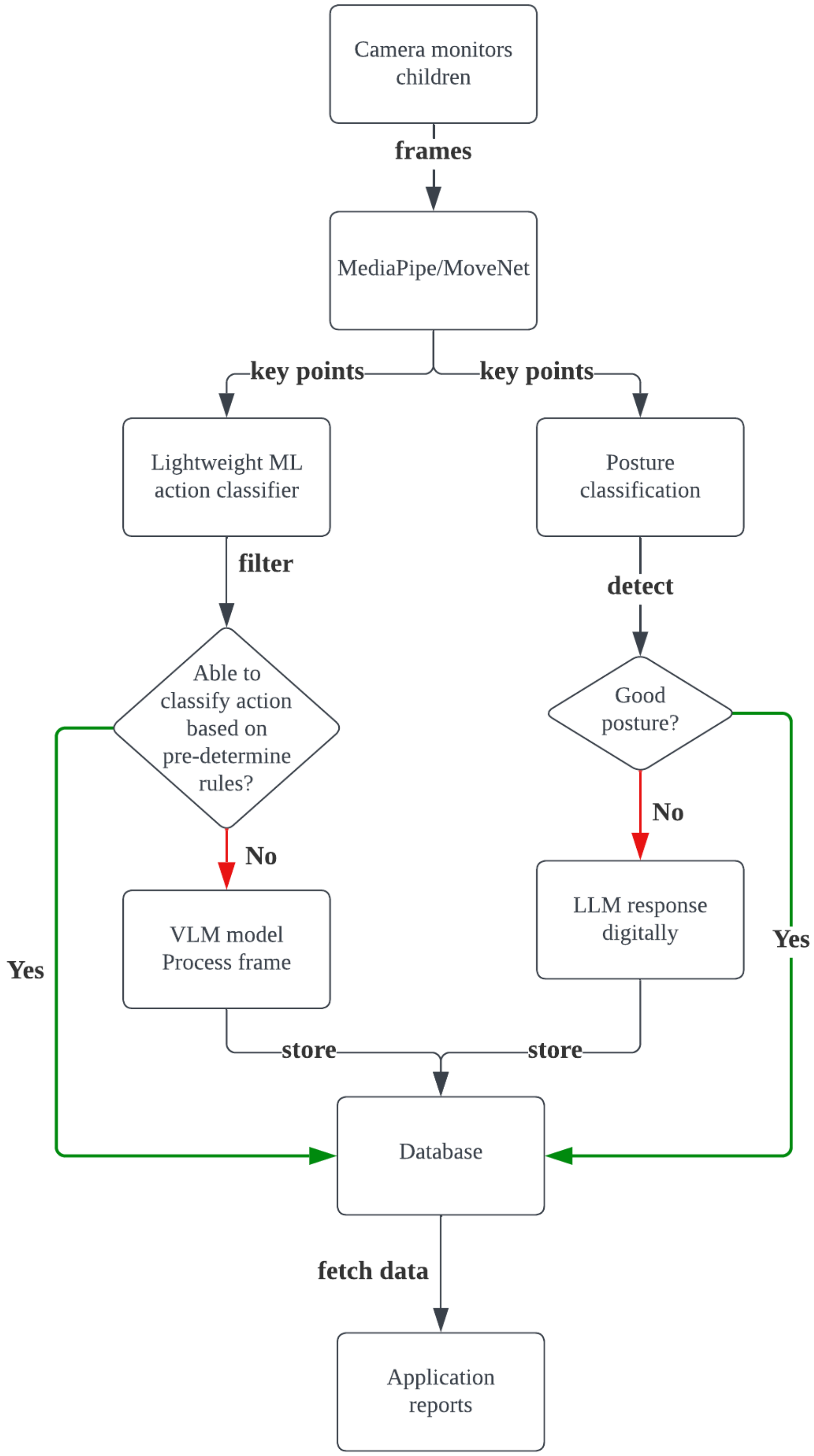
To create Monitra, the current idea involves integration of multiple AI models, each with their specific task. In the case of monitoring children's activity, Visual-Language Model is proposed to be used here instead of traditional Computer Vision, CV, due to its ability to understand text and image input, allowing for better understanding of context of frames. VLM, are Large Language Model, that has a visual encoder, allowing the AI model to 'see' and understand context of video/pictures[15].

Lightweight models to classify frames will have rules that are hardcoded, to determine obvious action according to the key points provided by MediaPipe or MoveNet. The reason is because certain actions are very easy to classify such as studying or not studying when looking at the key points relationships, however lightweight machine learning are not always accurate and are not reliable outside the hardcoded rules, hence frames that fall outside of the obvious rules that are pre-determined will be passed to VLM for a deeper and better understanding. Hence, this lightweight model will be using **Python** for any AI / ML purposes in Visual Studio Code.

Meanwhile, for posture monitoring, it is very challenging to only use VLM for this task, and the current choice are **MediaPipe** or **MoveNet**. The reason behind these 2 choices is that they are lightweight models that can be implemented on a user's device without **using too many resources** at the same time providing posture detection. To remind children of their posture, technology such as LLM will be utilized due to its ability to generate natural, conversational text that is not hard coded, promoting user-friendly reminders that are unique and interesting.

Additionally, to inform parents of related information, technology to create mobile applications include **Flutter**, and API that connects the mobile application to the webapp will be using **Flask**. The current reason behind these choices is because Flutter is easy to develop and has great UI, and the potential of cross-platform for further development if needed, enabling the practice of modularity.

4.2.3 Flow of Monitra



1. **Camera monitors children**

In this process, it refers to the separate camera that will be used to monitor children from an angle. The video stream would then be sent over to the web app for frames processing.

2. **MediaPipe/MoveNet**

Frames obtained from video stream will be applied to models such as MediaPipe or MoveNet for key extraction, that shows the key points of human body, such as joints and other parts of the body.

3. **Lightweight ML action classifier**

Refers to a model that will have pre-determined rules that classify actions into ‘Studying’ or ‘Not Studying’ based on hardcoded key-points. If the rules are not met, indicating a more complex action, frames will then be passed to VLM for processing.

4. **Posture detection**

By using key points extracted, rules will be created to check for posture and classify into ‘Good’ or ‘Bad’ posture.

5. **Classifying action (Diamond block)**

Based on the results of previous process of lightweight action classifier, results will either be stored in Database or passing frames to VLM for further processing.

6. **Classifying posture (Diamond block)**

Based on the results obtained from posture detection, ‘Bad’ posture will trigger the process of ‘LLM Response’, and either result must be store in Database as well.

7. VLM process frame

A VLM model will be chosen for this task, and process frames that was not able to be identified by lightweight action classifier.

8. LLM digital response

Utilising LLM for a digital response.

9. Database

A place to store information of processed data from this system

10. Application Reports

Mobile application made for parents to help inform them will need to access data from database to create reliable reports or dashboards.

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IIPSPW Report Marking Sheet

Mark Allocation	Key Assessment	Marks
10.0%	**Project Background** - What is the problem domain of this project? - Who has the problem and need a solution? - Why is the problem so important?	
10.0%	**Literature Review** - What have other researchers/developers done to resolve the problem? - What are the strengths of their solutions? - What are the weaknesses/limitations of their solutions? - How these weaknesses/limitations can be resolved?	
10.0%	**Project Scope and Objectives** - Is the title clear and specific enough? - What is the proposed solution of this project to tackle the above problem/limitations? - What is the scope? - What is the main objective, and how this objective can be divided into sub-objectives of the project? - What is the innovation/contribution of the project?	
5.0%	**Methods/Technologies Involved** - What are the proposed methods/technologies involved in the solutions? - How to justify the objective/sub-objectives can be achieved with these methods/technologies	
5.0%	**Report Format / Grammar / References** - Are the references adequate? - Is the report free from grammatical and writing errors?	
40.0%	**Total**	